Biochemistry of dietary, functional and structural lipids I, II

1- What is the primary structural distinction that categorizes lipid derivatives (such as eicosanoids and ketone bodies) within the comprehensive classification of lipids?

A) They are synthesized exclusively from protein backbones conjugated with carbohydrate groups

B) They are hydrolysis products of simple or compound lipids that still retain general lipid characteristics

C) They contain multi-ring steroid nuclei without any aliphatic side chains

D) They consist strictly of monohydric waxes lacking any ester bonds

ANSWER: B

2- A biochemical toxicologist is analyzing the mechanism of a snake venom component that exhibits strong hemolytic activity on red blood cells. The enzyme acts directly on lecithin by cleaving a fatty acid ester bond, converting lecithin into lysolecithin. Which of the following enzymes is responsible for this reaction?

A) Phospholipase C

B) Phospholipase A2

C) Lipoprotein lipase

D) Hormone-sensitive lipase

ANSWER: B

3- A premature neonate born at 28 weeks gestation develops severe labored breathing, cyanosis, and progressive hypoxia shortly after birth. Chest radiography reveals diffuse atelectasis (lung collapse). Which specific pulmonary substance is directly responsible for this condition?

A) Lysolecithin

B) Sphingomyelin

C) Phosphatidylserine

D) Dipalmitoyl phosphatidylcholine

ANSWER: D

4- A 28-year-old strict vegan patient exhibits dry scaly skin lesions, poor resistance to bacterial infections, and mild fatty liver changes on ultrasound. Laboratory tests confirm an essential fatty acid deficiency. The lack of which specific precursor fatty acid is the primary metabolic root of this clinical presentation?

A) Oleic acid

B) Palmitic acid

C) Linoleic acid

D) Stearic acid

ANSWER: C

5- Which of the following represents a volatile short-chain saturated fatty acid?

A) Palmitic acid

B) Stearic acid

C) Butyric acid

D) Arachidic acid

ANSWER: C

6- Apoprotein B-48 is uniquely synthesized and assembled into which of the following lipoproteins within the small intestine?

A) Chylomicrons

B) VLDL

C) LDL

D) HDL

ANSWER: A

7- Which of the following fatty acids belongs to the omega-3 family?

A) Linoleic acid

B) Arachidonic acid

C) α-Linolenic acid

D) Palmitoleic acid

ANSWER: C

8- Ceramide is formed by the combination of sphingosine and a fatty acid linked via which of the following chemical bonds?

A) Ester bond

B) Amide bond

C) Ether bond

D) Glycosidic bond

ANSWER: B

9- Phosphatidylcholine is more commonly known by which of the following names?

A) Cephalin

B) Sphingomyelin

C) Lecithin

D) Plasmalogen

ANSWER: C

10- Which of the following is known as "bad cholesterol" due to its role in increasing the risk of atherosclerosis?

A) Chylomicrons

B) High-Density Lipoproteins (HDL)

C) Low-Density Lipoproteins (LDL)

D) Intermediate-Density Lipoproteins (IDL)

ANSWER: C

11- Which of the following is the direct precursor for the synthesis of steroid hormones and vitamin D3?

A) Glycerol

B) Cholesterol

C) Lecithin

D) Sphingosine

ANSWER: B

12- Which of the following describes the chemical nature of glycerol?

A) A monohydric long-chain alcohol

B) A sterol containing four fused rings

C) A trihydric water-soluble alcohol

D) An amino alcohol containing 18-carbon chain

ANSWER: C

13- Which of the following is classified as a simple lipid?

A) Phosphatidylcholine

B) Triacylglycerol

C) Sphingomyelin

D) Lipoprotein

ANSWER: B

14- Which of the following is the main linkage joining alcohol and fatty acids together to form lipids?

A) Peptide bond

B) Amide bond

C) Ester bond

D) Phosphodiester bond

ANSWER: C

15- Which of the following lipids provides approximately 9 kcal of energy per gram?

A) Proteins

B) Carbohydrates

C) Nucleic acids

D) Lipids

ANSWER: D

16- Which of the following fatty acids contains one double bond?

A) Stearic acid

B) Palmitic acid

C) Oleic acid

D) Arachidic acid

ANSWER: C

17- Which of the following phospholipids is the most abundant phospholipid in cell membranes?

A) Lecithin

B) Cephalin

C) Sphingomyelin

D) Phosphatidic acid

ANSWER: A

18- Which of the following phospholipids plays an important role in blood clotting?

A) Sphingomyelin

B) Cephalin

C) Lecithin

D) Cholesterol

ANSWER: B

19- Which of the following structural components forms the core unit of all sphingolipids?

A) Lecithin

B) Ceramide

C) Cholesterol ester

D) Diacylglycerol

ANSWER: B

20- A patient with peripheral neuropathy is found to have degeneration of the myelin sheath surrounding nerve fibers. Which of the following phospholipids is normally abundant in the myelin sheath?

A) Lecithin

B) Phosphatidic acid

C) Sphingomyelin

D) Phosphatidylinositol

ANSWER: C